

**RAMCO INSTITUTE OF TECHNOLOGY**  
**Department of Electronics and Communication Engineering**  
**Academic Year: 2019- 2020 (Odd Semester)**

**Innovative Practices Description**

**UNIT I DISCRETE FOURIER TRANSFORM**

**Degree, Semester& Branch: V Semester B.E. ECE A**

**Course Code & Title: EC8553 Discrete Time Signal Processing**

**Name of the Faculty member: Mrs.S.Jeeva**

**Name of the Activity: Multimedia Approach & Concept Map**

**Name of the Topic: Introduction to digital signal processing & Overview of Syllabus**

**Date & Duration: 02.07.2019 & 10 Minutes**

**Description:**

Multimedia is characterized by the presence of text, pictures, sound, animation and video; some or all of which are organized into some coherent program. Multimedia approach is an approach of teaching in which different mediums are incorporated to make the teaching-learning more effective, enthusiastic. The highly presence and motivation also bring positive aspects to students so that they can improve their skills.

**Goal (Learning Outcomes):**

- The students will be able to know the importance & applications of digital signal processing in various fields.

**Use of appropriate method:**

**Justification for choosing the Multimedia approach activity for this topic:**

Use of Multimedia (audio-visual aids) help in maintaining discipline in the class since all the students' attention are focused in learning. This interactive session also develops critical thinking and reasoning that are important components of the teaching-learning process. Students learn when they are motivated and curious about something. Traditional verbal instructions can be boring and painful for students. However, use of audio-visual provides intrinsic motivation to students by peaking their curiosity and stimulating their interests in the subjects.

**Effective Presentation:**

**Implementation (Plan & Execution) with Proof:**

- Beginning of this semester I shown video about fundamentals of digital signal processing, applications of digital signal processing.
- I shown flow chart about overview of syllabus.



### **Reflective Critique:**

#### **Benefits:**

Due to this activity students can able to

- construct knowledge about signals, systems, digital signal processing
- solve problems related systems
- Students are able to understand the basics of signals & systems, importance of digital signal processing in real word application.

#### **Challenges:**

- Need to select best Video selection with shorter delivery duration with clear audio.

#### **Name of the Activity: One Minute Paper**

**Name of the Topic:** DIT,DIF Algorithm

**Date & Duration:** 12.07.2019 & 5 Minutes

#### **Description:**

The one-minute paper is a versatile and easily employable assessment technique.It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. It is a short writing activity within the classroom in which students generate answers in response to questions asked by the teacher.

#### **Goal (Learning Outcomes):**

- The students will be able to remember DIT, DIF radix-2 butterfly diagram.

#### **Use of appropriate method:**

#### **Justification for choosing the One minute paper activity for this topic:**

DIT, DIF radix-2 algorithm is an important topic in concept wise and university exam point of view. Students are must be known butterfly diagram due to this activity the students are got practice to draw diagram correctly.

**Effective Presentation:****Implementation (Plan & Execution) with Proof:**

Students are asked to take bit of paper & give 1 minute to draw basic butterfly flow graph of DIT & DIF algorithm. After 1 minute collect their answer scripts and discuss about correct solution.

**Reflective Critique:****Benefits:**

- Due to this activity students are able to recall the basic butterfly structure of algorithm, it will be more helpful to revise internal assessment test I portion.

**Challenges:**

- Totally I Planned to conduct this activity for 5 minutes but it took around 10 minutes.

**Significance of Results:****Assessment of effectiveness/Success of the activity:**

- With audio & video presentation the students got idea about different real time digital applications.
- Due to this activities, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam.

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1            | 2   | 2   | 1   | 1   | 2   |     |     |     |     | 1    |      | 1    |

*CO1: The students will be able to Apply DFT for the analysis of digital signals & systems.*

**References:**

1. John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
2. Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.

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**UNIT II INFINITE IMPULSE RESPONSE FILTER**

**Degree, Semester& Branch: V Semester B.E. ECE A**

**Course Code & Title: EC8553 Discrete Time Signal Processing**

**Name of the Faculty member: Mrs.S.Jeeva**

**Name of the Activity: Think Pair Share**

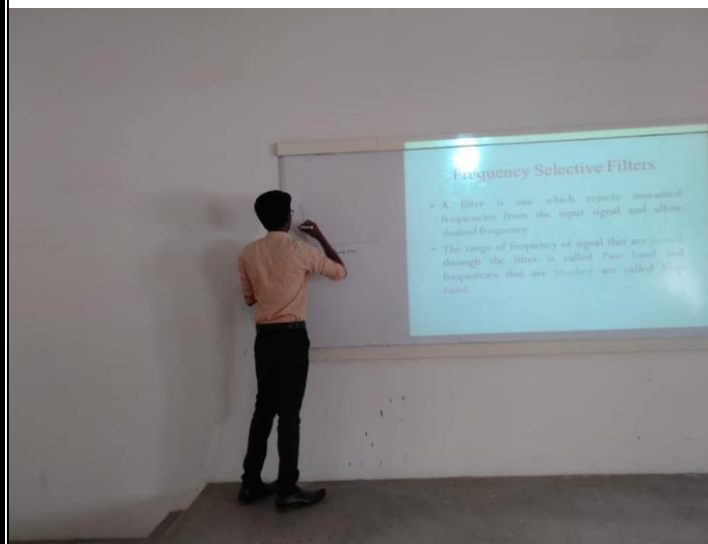
**Name of the Topic: Characteristics of frequency selective filters**

**Date & Duration: 18.07.2019 & 10 Minutes**

**Description:**

This technique gives students the opportunity to respond to questions in written form before engaging in meaningful conversation with other students. Asking students to write and discuss ideas with a partner before sharing with the larger group builds confidence, encourages greater participation, and results in more thoughtful discussions.

Think-Pair-Share (TPS) is a collaborative learning strategy where students collectively work together to solve a problem or answer to a reading assignment.



**Goal (Learning Outcomes):**

1. The students will be able to recollect the characteristics of frequency selective(LP,HP,BP,BR)filters.
2. The students will be able to know the various method of digital IIR filter design techniques.

### **Effective Presentation:**

#### **Implementation(Plan & Execution) with Proof:**

1. **T** : (Think) First I'm asked question about characteristics of frequency selective filters. Students "think" about what they know or have learned about the topic.
2. **P** : (Pair) Each student should be paired with another student.
3. **S** : (Share) Students share their thinking with their partner. Finally I expand the "share" into a whole-class discussion.

### **Use of appropriate Method:**

#### **Justification for choosing the Think Pair Share activity for this topic:**

- Students are already know about the characteristics of filters in their 3<sup>rd</sup> semester. So this activity is used to refresh the known concepts.

### **Reflective Critique:**

#### **Benefits:**

- ✓ Due to this activity the students are able to recollect the points of filters
- ✓ Improve their remembering capacity

#### **Challenges:**

- Some of the students are not able to recollect all the points
- Most of the points are repeated
- It takes additional 5 minutes to complete that activity compared to planned duration

#### **Name of the Innovative Practice: Flash Card**

**Date & Duration: 26.07.2019 & 10 Minutes**

#### **Name of the Topic: Digital IIR filter design techniques**

#### **Description:**

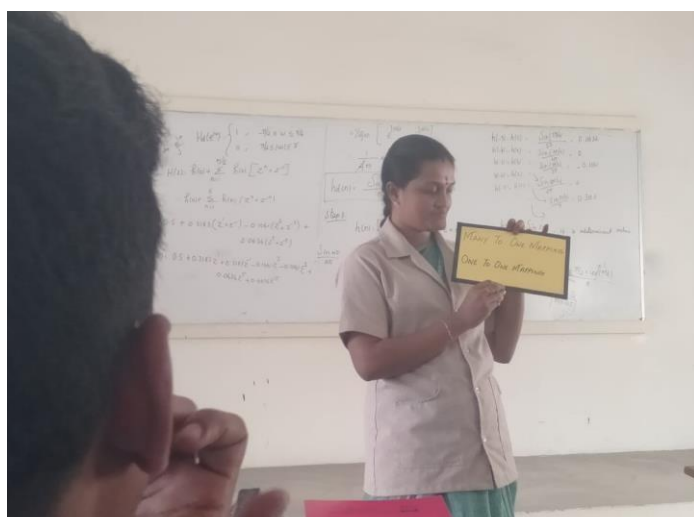
Flash card is a card bearing information on both sides, which is intended to be used as an aid in memorization. Each flashcard bears a question on one side and an answer on the other. Flashcards are often used to memorize vocabulary, historical dates, formulae or any subject matter that can be learned via a question-and-answer format. Flashcards can be virtual (part of a flashcard software), or physical.

Flashcards are an application of the testing effect – the finding that long-term memory is increased when some of the learning period is devoted to retrieving the information through testing with proper feedback.

### **Effective Presentation:**

#### **Implementation (Plan & Execution) with Proof:**

- First I will write digital IIR filter techniques in one side of chart and write another name of this technique in other side of this chart.



### **Use of appropriate Method:**

#### **Justification for choosing the flash card activity for this topic:**

- Design of digital IIR has various methods. It is a easy way of remember the different names of digital IIR filter design techniques

### **Significance of Results:**

#### **Assessment of effectiveness/Success of the activity:**

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test-I most of the students attended the questions under this topic and scored good marks.

### **Reflective Critique:**

#### **Benefits:**

- ✓ Due to this activity students memory capacity would be improved & they will understand the fundamental concept of filters, design techniques
- ✓ Easily revised the previous class topic

#### **Challenges:**

- Need to prepare more number of flash cards will be prepared.

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO2            | 2   | 2   | 1   | 1   | 1   |     |     |     |     | 1    |      | 1    |

### **References:**

1. Andreas Antoniou, 'Digital Signal Processing', Tata McGraw Hill, 2006.
2. Nagoor Kani, 'Digital Signal Processing', Tata McGraw Hill, Second Edition.
3. Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.

*CO2: The Students will be able to design Infinite Impulse response (IIR) digital filters.*

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**UNIT III FINITE IMPULSE RESPONSE FILTERS**

**Degree, Semester& Branch: V Semester B.E. ECE A**

**Course Code & Title: EC8553 Discrete Time Signal Processing**

**Name of the Faculty member: Mrs.S.Jeeva**

**Name of the Activity: Exit slip**

**Date & Duration: 21.08.2019 & 10 Minutes**

**Name of the Topic: Windowing Methods**

**Description:**

**Exit slips** are written student responses to questions teachers pose at the end of a class or lesson. These quick, informal assessments enable teachers to quickly assess students' understanding of the material.

**Goal (Learning Outcome):**

- The Students will be able to recollect the points at the end of class
- The students can improve their memory capacity for small writeup activity

**Use of appropriate method:**

**Justification for choosing the Exit slip activity for this topic:**

- More number of formulae are used for windowing method so students are must practice to write correctly. Due to this activity the students are have the habit of writing the formulae.

**Effective Presentation:**

**Implementation (Plan & Execution) with Proof:**

- At the end of class I will post the questions to students, they will think and write appropriate answer in paper.



Fig 1: Students write answers

|                                         |                                                                                                                                               |                                                                                                                                                |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Step 1                                  | Filter coefficient                                                                                                                            | Filter coefficient                                                                                                                             |
| Step 2                                  | Magnitude & Phase Response                                                                                                                    | Magnitude & Phase Response                                                                                                                     |
| Window sequences for FIR filter design: |                                                                                                                                               |                                                                                                                                                |
| Window                                  | Window sequence                                                                                                                               |                                                                                                                                                |
| Rectangular                             | $w_p(n) = \begin{cases} 1 & -\frac{(N-1)}{2} \leq n \leq \frac{(N-1)}{2} \\ 0 & \text{otherwise} \end{cases}$                                 |                                                                                                                                                |
| Hanning                                 | $w_p(n) = \begin{cases} 0.5 + 0.5 \cos \frac{2\pi n}{N-1} & -\frac{(N-1)}{2} \leq n \leq \frac{(N-1)}{2} \\ 0 & \text{otherwise} \end{cases}$ |                                                                                                                                                |
| Hanning                                 | $w_p(n) = \begin{cases} 0.5 + 0.5 \cos \frac{2\pi n}{N-1} & -\frac{(N-1)}{2} \leq n \leq \frac{(N-1)}{2} \\ 0 & \text{otherwise} \end{cases}$ |                                                                                                                                                |
| To find Filter Type:                    |                                                                                                                                               |                                                                                                                                                |
| Type Filter                             | Diagram                                                                                                                                       | Ideal filter Response                                                                                                                          |
| Low Pass Filter                         |                                                                                                                                               | $H_d(e^{j\omega}) = \begin{cases} 1 & -W_c \leq \omega \leq W_c \\ 0 & \text{otherwise} \end{cases}$                                           |
| High Pass Filter                        |                                                                                                                                               | $H_d(e^{j\omega}) = \begin{cases} 1 & -W_{c1} \leq \omega \leq W_{c2} \\ 0 & \text{otherwise} \end{cases}$                                     |
| Band Pass Filter                        |                                                                                                                                               | $H_d(e^{j\omega}) = \begin{cases} 1 & W_{c1} \leq \omega \leq W_{c2} \\ 0 & \text{otherwise} \end{cases}$                                      |
| Band Stop Filter                        |                                                                                                                                               | $H_d(e^{j\omega}) = \begin{cases} 1 & -W_{c1} \leq \omega \leq W_{c1} \\ & W_{c2} \leq \omega \leq W_{c2} \\ 0 & \text{otherwise} \end{cases}$ |

|                                 |                                                                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Discrete-Time Signal Processing |                                                                                                                                                 |
| Window function formula:        | Regno: 95367106024<br>17/04/2024                                                                                                                |
| i) Rectangular window:          |                                                                                                                                                 |
| Type-I =                        | $w_p(n) = \begin{cases} 1 & -\frac{(N-1)}{2} \leq n \leq \frac{(N-1)}{2} \\ 0 & \text{otherwise} \end{cases}$                                   |
| Type-II =                       | $w_p(n) = \begin{cases} 1 & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$                                                              |
| ii) Hanning window:             |                                                                                                                                                 |
| Type-I =                        | $w_p(n) = \begin{cases} 0.5 + 0.5 \cos \frac{2\pi n}{N-1} & -\frac{(N-1)}{2} \leq n \leq \frac{(N-1)}{2} \\ 0 & \text{otherwise} \end{cases}$   |
| Type-II =                       | $w_p(n) = \begin{cases} 0.5 + 0.5 \cos \frac{2\pi n}{N-1} & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$                              |
| iii) Hamming window:            |                                                                                                                                                 |
| Type-I =                        | $w_p(n) = \begin{cases} 0.54 + 0.46 \cos \frac{2\pi n}{N-1} & -\frac{(N-1)}{2} \leq n \leq \frac{(N-1)}{2} \\ 0 & \text{otherwise} \end{cases}$ |
| Type-II =                       | $w_p(n) = \begin{cases} 0.54 + 0.46 \cos \frac{2\pi n}{N-1} & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$                            |

Fig 2: Sample Exit slip

### Reflective Critique:

#### **Benefits:**

- ✓ The students are eagerly write the answers with fast remembering capacity

#### **Challenges:**

- Actually I planned for 10 minutes but it takes extra 5 minutes



**Name of the Activity: Mindmap**

**Date & Duration: 29.08.2019**

**Name of the Topic: FIR Filters**

**Description:**

Mind mapping is a visual form of note taking that offers an overview of a topic and its complex information, allowing students to comprehend, create new ideas and build connections.

**Goal (Learning Outcomes):**

- The students will be able to think different ideas to remember the concepts
- The students improve their self learning skills

**Use of appropriate method:**

**Justification for choosing the Mind map activity for this topic:**

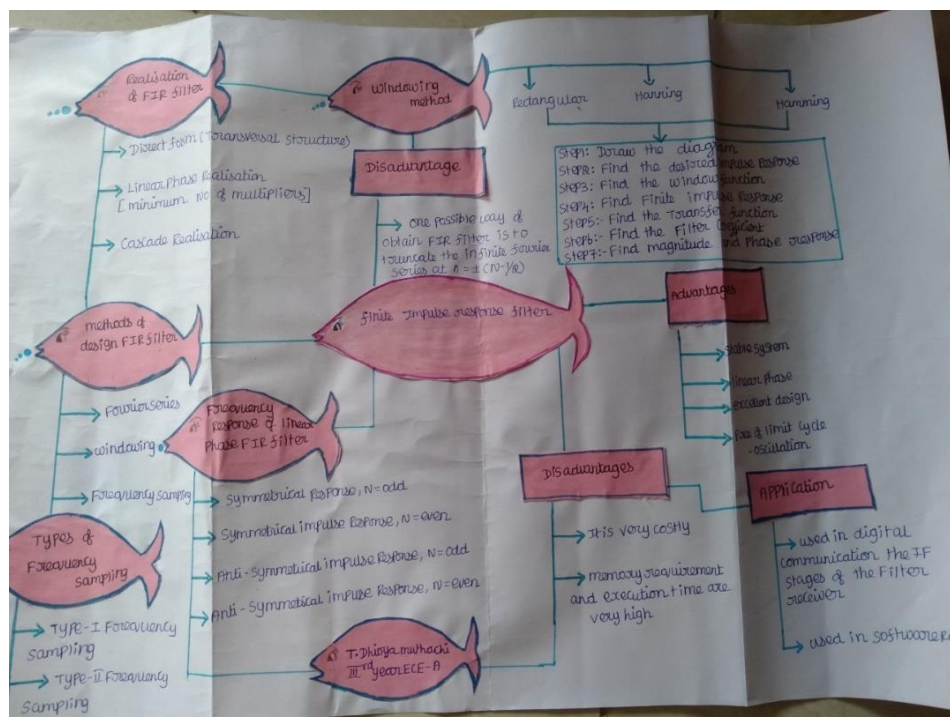
- Mind map is like a flow chart, so students are easily understand the design techniques of Filter design
- This activity increases students creativity
- It is a simple way of revise the topics

**Effective Presentation:**

**Implementation (Plan & Execution) with Proof:**

- Two days before, I give the Mind map activity topic to the students.
- At the time of activity the volunteers they will come forward and present their flow chart to others.





**Fig: Sample Mind Map**

### **Significance of Results:**

#### **Assessment of effectiveness/Success of the activity:**

- Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test-II most of the students attended the questions under this topic and scored good marks.

### **Reflective Critique:**

#### **Benefits:**

- ✓ Due to this activity the students are felt easy to understand the topics
- ✓ Students are actively participate and share their ideas in creative manner

#### **Challenges:**

- Some of the students are difficult to recollect all the formula related to filter design

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO3            | 2   | 2   | 2   | 1   | 1   |     |     | 1   |     | 3    |      | 1    |

*CO3: The Students will be able to design Finite Impulse response (FIR) digital filters.*

#### **References:**

1. S.K. Mitra, 'Digital Signal Processing – A Computer Based Approach', McGraw Hill Edu, 2013.
2. Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.
3. Nagoor Kani, 'Digital Signal Processing', Tata McGraw Hill, Second Edition

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**UNIT IV FINITE WORDLENGTH EFFECTS**

**Degree, Semester& Branch: V Semester B.E. ECE A**

**Course Code & Title: EC8553 Discrete Time Signal Processing**

**Name of the Faculty member: Mrs.S.Jeeva**

**Name of the Activity: Crossword Puzzle**

**Date & Duration: 17.09.2019 & 15 Minutes**

**Name of the Topic: Finite Wordlength Effects**

**URL: <http://www.classtools.net/crossword/201910-dmHXVF>**

**Description:**

A crossword is an interesting word puzzle activity which helps the students to recollect the concept. Crossword puzzles provide students with an opportunity to evaluate their level of knowledge and to identify areas of further study in a fun way. Hence, including crosswords in the Engineering field teaching repertoire can be an innovative way to engage students in learning and help them to acquire knowledge and skills. It is one type of game, so this an enjoyable activity rather than regular multiple choice questions test.

**Goal (Learning Outcome):**

- The students are able to remember the concepts at the time of examination
- The students are got confidence to study those topics properly

**Use of appropriate Method:**

**Justification for choosing the Crossword Puzzle activity for this topic:**

- This topic is like full theory & many type of errors so students are must read all content given in prescribed book
- Due to this activity the students are involved in text book reading & find out the answer

**Effective Presentation:**

**Implementation (Plan & Execution) with Proof:**

- ✓ First of all I am generate crossword puzzle by using classtool.net, it consists of 10 questions (Across-5 questions, Down-5 questions)
- ✓ The students are divided into groups (Boys-4 per team, Girls-6 per team)

- ✓ Individual Crossword puzzle sheets are provided to each group.
- ✓ The students answered the questions in the first 20 minutes of the class.
- ✓ The answer scripts are collected and evaluated by other group members.
- ✓ Finally I discussed the answers of the wrong questions at the end of the class.



### **Reflective Critique:**

#### **Benefits:**

- ✓ Due to this activity the students should be able to recall information easily and improve their collaborative learning skills.

#### **Challenges:**

- Some of the students are difficult to search the answer from text book
- Actually planned within 15 minutes but it takes 20 Minutes (5 minutes delay)

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO4            | 2   | 2   | 2   | 1   |     |     |     |     |     | 2    |      | 1    |

### **References:**

1. Andreas Antoniou, 'Digital Signal Processing', Tata McGraw Hill, 2006.
2. Ramesh Babu, 'Digital Signal Processing', SciTech Publications.

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**UNIT V INTRODUCTION TO DIGITAL SIGNAL PROCSSORS**

**Degree, Semester& Branch: V Semester B.E. ECE A**

**Course Code & Title: EC8553 Discrete Time Signal Processing**

**Name of the Faculty member: Mrs.S.Jeeva**

**Name of the Activity: Flipped Classroom**

**Date & Duration: 11.10.2019 & 20 Minutes**

**Name of the Topic: Applications of DSP Processor**

**Description:**

Flipped Classroom is a pedagogical approach in which the conventional notion of classroom based learning is inverted, so that students are introduced to the learning material before class, with classroom time then being used to deepen understanding through discussion with peers and problem solving activities.

**Goal (Learning Outcomes):**

At the end of the flipped class activity,

- ❖ The students will be able to explain the various applications of DSP Processors with real time examples in the areas of medical, signal & image processing, communication systems.
- ❖ The self learning skill and team work skills will be enhanced.

**Use of appropriate method:**

**Justification for Choosing the Topic using Flipped Class:**

The topic was chosen for flipped class as there are many applications for Digital Signal Processors. The Unit 5 deals with the details of digital signal processor architecture, functionalities, fixed and floating point processors, instruction sets, programming and finally application examples. Since the students are exposed to the details of processor, I thought of giving the topic '*Application of Digital Signal Processors*' for flipped class. This topic can be learnt in self-learning mode.

- ❖ As there are many applications of Digital signal processors, only 10 applications in various domains were discussed in their prescribed text book. If this flipped class is used for the topic apart from the learning materials given by the faculty, students have chance to browse and find more information related to specific application. Thus 10

applications of Digital Signal Processor will be covered in 2 periods through this activity which is not possible in traditional teaching. This is the reason for choosing this topic for flipped class activity.

**Effective Presentation:**

**Implementation (Plan & Execution) with Proof:**

- ✓ The students are divided into groups (Boys-4 per team, Girls-6 per team). A group should be mixer of advance learners, average learners and slow learners.
- ✓ The students were informed about the flipped class activity atleast 2 weeks before the implementation of the activity.
- ✓ The students have to prepare a presentation for 10 minutes with the concept learnt and 3 minutes for Question & answering and tell about the contribution of each member.
- ✓ The inside class activity: presentation will be planned for 2 periods. In which each group has to present their topic chosen and others will also listen to the presentation.
- ✓ The applications of digital signal processors are covered to the whole class and the handouts prepared on the topic will be shared within the group members.
- ✓ Each group will be assessed through the content of presentation (20 Marks), communication skills (10 Marks), Q& A (10 Marks), contribution by each member (10 Marks).



**Reflective Critique:**

**Benefits:**

- ✓ The students had good experience of learning a topic out the class and presentation in the class.
- ✓ Since the topic is related to application, they felt very interesting to know the real time applications of Digital Signal Processors.
- ✓ The Activity was conducted for groups, Interactions and discussions which collaborations made them to learn more information.



**Challenges:**

- Making all the students to involve in discussion thread initially.
- Time management – Since few group of students made lengthy presentation, it was difficult to complete in 10 minutes.
- In Q& A module – few students struggled to answer the questions and ask question to the peers.
- Few students had stage fear while presentation.

**Significance of Results:****Assessment of effectiveness/Success of the activity:**

- Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test-III most of the students attended the questions under this topic and scored good marks.

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|----------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO5            | 2   | 2   | 2   | 1   |     |     |     | 1   |     | 2    |      |      |

*CO5: The students will be able to Explain the functionalities and architecture of DSP processors.*

**References:**

1. Venkatarami.B, Bhaskar.M, 'Digital Signal Processors, Architecture, programming & Applications', Tata McGraw Hill, Second Edition.
2. SenM.kuo, woonseng...s.gan, 'Digital Signal Processors, Architecture, Implementations &Applications', Pearson,2013